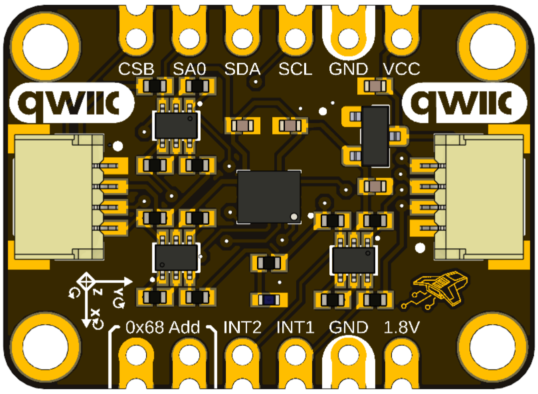
I2C BOSCH BMI323 IMU
DEVLAB FACTOR
Mfr. Part #: UE0132

Product Reference Manual (V1.0)

Description

This module features the **BOSCH BMI323,** a high-performance, and highly integrated low power inertial sensor module that integrates a triaxial accelerometer and a triaxial gyroscope, with a digital temperature sensor.It provides accurate measurement of linear acceleration and angular rate, making it well suited for motion sensing applications with low power requirements.



The module is designed for easy integration with microcontrollers and embedded systems, supporting **I²C interface** ensures easy communication and SPI allows for higher data transmission. Implements intelligent on-chip motion-triggered through the two configurable interrupt pins. Its small form factor and efficient power management make it ideal for wearables, IoT devices, and motion detection systems.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C and SPI interfaces.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.

This design guarantees efficient prototyping, accessibility, and simplicity of integration across a variety of devices and modules.

Key Features

- Bosch BMI323 6-axis Inertial Measurement Unit (IMU)
- 16-bit triaxial accelerometer and 16-bit triaxial gyroscope
- Selectable accelerometer ranges: ±2 g, ±4 g, ±8 g, ±16 g
- Selectable gyroscope ranges: ±125 dps, ±250 dps, ±500 dps, ±1000 dps, ±2000 dps
- Output Data Rate (ODR) up to 6.4 kHz
- Integrated digital temperature sensor
- Programmable digital filtering and averaging
- Integrated FIFO buffer for sensor and timestamp data
- Dual communication interfaces: I²C (Fast-mode Plus) and SPI
- Two programmable interrupt outputs pins(INT1 and INT2)
- Low-power operation with multiple power modes
- Hardware synchronization support
- DevLab-compatible form factor with Qwiic® and STEMMA QT connectors
- 3.3 V and 5 V logic compatibility through onboard level shifting
- Castellated pads for permanent PCB integration

Hardware Features

- Bosch BMI323 MEMS inertial sensor
- Integrated 3-axis accelerometer
- Integrated 3-axis gyroscope
- Integrated digital temperature sensor
- Onboard bidirectional logic level shifter
- Dual Qwiic® / STEMMA QT I²C connectors
- JST connector for SPI communication
- Castellated edge pads for direct soldering
- Two programmable interrupt pins (INT1 and INT2)
- Power and communication status designed for embedded applications
- Compact DevLab module form factor
- RoHS compliant design

Software Support

- Official and community-supported drivers available
- Supported environments include:
    - Arduino IDE
    - PlatformIO
    - ESP-IDF
    - MicroPython
    - CircuitPython
    - Raspberry Pi Pico SDK
    - STM32 HAL / LL
    - Zephyr RTOS
- Register-level configuration for full control of sensor features
- Built-in motion detection and activity recognition functions
- Configurable filters, output data rates, and power modes

Applications

- Wearable electronics
- Fitness and activity tracking
- Robotics and autonomous platforms
- Motion control systems
- Gesture recognition
- Human-machine interfaces (HMI)
- IoT sensor nodes
- Drone stabilization and navigation
- Industrial monitoring
- Vibration analysis
- Tilt and inclination measurement
- Step detection and activity recognition
- Asset tracking
- Smart home devices
- Educational and research platforms
- Rapid prototyping of embedded systems

CONTENTS

1 The Board

1.1 Accessories

- 2×6-pin 2.54 mm male header
- 1x QWIIC 4-pin JST 1.0 mm Harness

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VCC	Module Input Supply Voltage	3.3	5.0	5.5	V
VDD	BMI323 Supply Voltage	1.71	1.8	3.6	V
VDDIO	BMI323 I/O Supply Voltage	1.2	1.8	3.6	V
TA	Operating Ambient Temperature	-40	—	+85	°C
fSCL	I²C Clock Frequency	—	—	1	MHz
fSCK	SPI Clock Frequency	—	—	10	MHz

3 Functional Overview

The DevLab UNIT I²C BOSCH BMI323 IMU is a high-sensitivity and highly integrated, low power IMU, that combines precise acceleration and gyroscope measurement with intelligent on-chip motion-triggered interrupt features compatible with I²C interface and SPI interface. The BMI323 IMU integrates:

- A 16 bit digital,triaxial accelerometer with range configurable +/- 2G,+/-4G,+/-8G,+/-16G
- A 16 bit digital, triaxial gyroscope with range configurable to +/-125°/s,+/-250°/s,+/-500°/s,+/-1000°/s,+/-2000°/s
- A 16 bit digital temperature sensor for an operating temperature range -40°C to +85 °C

It integrates an onboard level shifter for 3.3 V and 5 V logic compatibility, ensuring reliable communication across a wide range of microcontrollers and development platforms.This sensor utilizes a standard two-wire I2C serial bus, via castellated holes or via either of the two onboard QWIIC connectors. And for the SPI serial bus, it can be connected via castellated holes, or via SPI JST connector at the bottom of the board(not included).

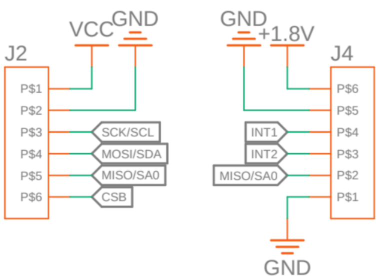
3.1 I²C Communication Interface

The module operates as a slave device on a standard I²C bus, using a fixed 7-bit address of **0x69** to establish all necessary communications and configuration tasks. Communication is fully bidirectional. Internally, to ensure proper functionality, the specific device identification register **(0x00)** can be read**; it returns a value of 0x43** in a 16 bit format , but only can read the LSB part, that is the first 8 bits, if the connection is successfully established.

3.1.1 Address 0x68

The device has two connectors in the castellated holes, with the label of 0x68 AD0, which you connect between these 2 pins, the I²C address to establish all necessary communications with the module will be 0x68.

Castellated Holes



3.2 SPI Communication Interface

The module also operates as a slave device on a standard SPI bus, using the same fixed 7-bit address that I²C Communication Interface, to establish all necessary communications and configuration tasks.

3.3 Configuration

The BMI323 DevLab Module provides different parameters to configure accelerometer and gyroscope.

For the accelerometer you have 5 modes of settings through the ACC_CONF address 0x20,which has output data rate, bandwidth, range, and the mode of the accelerometer.

The first setting is the ODR which has the first 4 bits of the address **[3:0]** which setting up the sample rate ODR in Hz for the measurement, it can be set or read through the same ACC_CONF, and the following table list all the frequencies for the ODR settings:

Field Value	Description
0x1	0.78125 Hz
0x2	1.5625 Hz
0x3	3.125 Hz
0x4	6.25 Hz
0x5	12.5 Hz
0x6	25 Hz
0x7	50 Hz
0x8	100 Hz
0x9	200 Hz
0xA	400 Hz
0xB	800 Hz
0xC	1.6 kHz
0xD	3.2 kHz
0xE	6.4 kHz

The second setting for the accelerometer is the range , that has 3 bits of the same ACC_CONF [6:4], this setting is for the configurable range, which can be set or read from this field, and the settings are listed in the following table:

Field Value	Description
0	+/- 2G, 16.38 LSB/mg
1	+/- 4G, 8.19 LSB/mg
2	+/- 8G, 4.10 LSB/mg
3	+/- 16G, 2.05 LSB/mg

The next setting is the bandwidth, with only 1 bit of configuration in the bit [7], which can be set or read from this field, and this configure the -3dB cut-off frequency for the accelerometer based on ACC_ODR, and are listed in the following table:

Field Value	Description
0	BW:ACC_ODR/2
1	BW:ACC_ODR/4

The fourth setting is the number of samples to be averaged to the samples that the sensor recollected of the measures, that has 3 bits in the field bits [10:8], and are listed in the following tables:

Field Value	Description
0	No averaging: pass simple without filtering
0x1	Averaging of 2 samples
0x2	Averaging of 4 samples
0x3	Averaging of 8 samples
0x4	Averaging of 16 samples
0x5	Averaging of 32 samples
0x6	Averaging of 64 samples

The last setting is the operation modes for the accelerometer, and it has 3 bits of configuration [14:12], in the same ACC_CONF, and the operation modes are the following:

Field Value	Description
0	Disables the accelerometer
0x3	Enables the accelerometer with sensing operated in duty-cycling
0x4	Enables the accelerometer in a continuous operation mode with reduced current
0x7	Enables the accelerometer in high performance mode

For the gyroscope you have the same 5 modes of settings through the GYR_CONF address 0x21, which has output data rate, bandwidth, range, and the mode of the gyroscope.

The first setting is the ODR which has the first 4 bits of the address **[3:0]** which setting up the sample rate ODR in Hz for the measurement, it can be set or read through the same GYR_CONF, and the following table list all the frequencies for the ODR settings:

Field Value	Description
0x1	0.78125 Hz
0x2	1.5625 Hz
0x3	3.125 Hz
0x4	6.25 Hz
0x5	12.5 Hz
0x6	25 Hz
0x7	50 Hz
0x8	100 Hz
0x9	200 Hz
0xA	400 Hz

0xB	800 Hz
0xC	1.6 kHz
0xD	3.2 kHz
0xE	6.4 kHz

The second setting for the accelerometer is the range , that has 3 bits of the same GYR_CONF [6:4], this setting is for the configurable range, which can be set or read from this field, and the settings are listed in the following table:

Field Value	Description
0	+/- 125°/s, 262.144 LSB/°/s
1	+/- 250°/s,, 131.072 LSB/°/s
2	+/- 500°/s,, 65.536 LSB/°/s
3	+/- 1000°/s,, 32.768 LSB/°/s
4	+/- 2000°/s, 16.4 LSB/°/s

The next setting is the bandwidth, with only 1 bit of configuration in the bit [7], which can be set or read from this field, and this configure the -3dB cut-off frequency for the accelerometer based on GYR_ODR, and are listed in the following table:

Field Value	Description
0	BW:GYR_ODR/2
1	BW:GYR_ODR/4

The fourth setting is the number of samples to be averaged to the samples that the sensor recollected of the measures, that has 3 bits in the field bits [10:8], and are listed in the following tables:

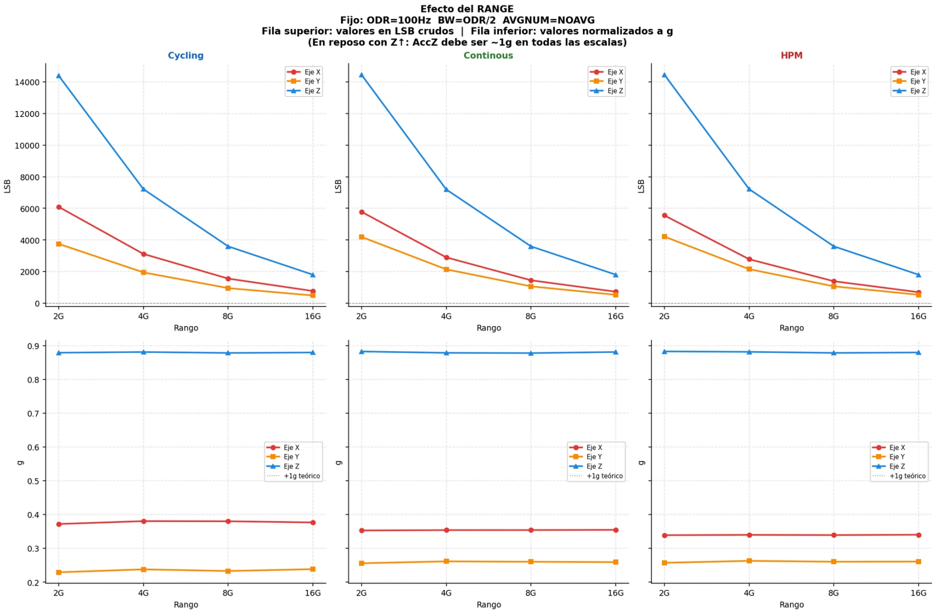
Field Value	Description
0	No averaging: pass simple without filtering
0x1	Averaging of 2 samples
0x2	Averaging of 4 samples
0x3	Averaging of 8 samples

0x4	Averaging of 16 samples
0x5	Averaging of 32 samples
0x6	Averaging of 64 samples

The last setting is the operation modes for the accelerometer, and it has 3 bits of configuration [14:12], in the same GYR_CONF, and the operation modes are the following:

Field Value	Description
0	Disables the accelerometer
0x1	Disables the gyroscope but keep the gyroscope drive enabled
0x3	Enables the gyroscope with sensing operated in duty cycling
0x4	Enables the gyroscope in a continuous operation mode with reduced current.
0x7	Enables the gyroscope in high performance mode

In the next picture, you can view one of the best settings for measures , with the 3 modes of settings for the accelerometer, in the superior row you can view the medium raw data of the measures, with the z axis static, which the z axis value is approximately 1 depends on the filter, measure and the settings. And in the second row you can view the value of the gravitational unit, measured at the Z axis.



3.4 Interrupt System

The module includes an interrupt system that supports two external pins, each configurable to map a custom interrupt depending on the feature engine.

Register **0x38** is used to configure and enable the physical interrupt pins (**INT1** and **INT2**):

- **Active Level:** Bit [0] sets **INT1** (0x0: active low, 0x1: active high), while Bit [8] sets **INT2** (0x0: active low, 0x1: active high).
- **Output Mode:** Bit [1] configures **INT1** (0: push-pull, 1: open-drain), and Bit [9] configures **INT2** (0: push-pull, 1: open-drain).
- **Interrupt Enable:** Bit [2] enables **INT1** when set to 1, and Bit [10] enables **INT2** when set to 1."

The clear behavior of the system is managed by the `INT_CONF` in the latch field (Bit 0). Setting this bit to **(0x0)** configures the system as **non-latched**, while setting it to **(0x1)** makes the interrupts **permanently latched**. To clear the physical state of the interrupt pins when latched, write to register address **0x39**.

Some interrupts require the feature engine to be configured; otherwise, they will not function.

To activate the feature engine, use the `enableFeatEngine` function from the **DevLab BMI323** Arduino library. This function automatically performs the following sequence:

- **Soft Reset**
- Write `0x012C` to `FEATURE_IO2` (0x12)
- Write `0x0001` to `FEATURE_IO_STATUS` (0x14)
- Write `0x1` to `FEATURE_CTRL` (0x40)

Important: Always enable the sensors *after* calling this function. Activating the feature engine while the sensors are already running will result in inaccurate measurements.

Finally, configure the interrupt mapping destination. Each physical pin (**INT1** and **INT2**) supports only one interrupt source at a time. The multiple interrupt settings are distributed across two different mapping registers. Register **0x3A** contains the following interrupt options:

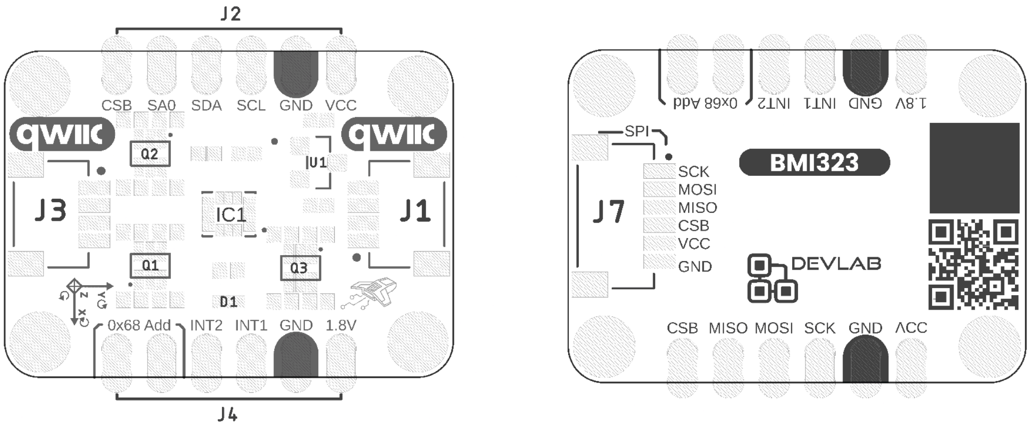
- **No Motion Out:** No motion, or no movement
- **Any Motion Out:** Any motion, or any movement captured.
- **Flat Out:** Sensor in flat position.
- **Orientation Out:** Orientation of the sensor
- **Step Detector Out:** Step or stride detected
- **Step Counter Out:** Target step count reached
- **Sig Motion Out:** Significant or sustained movement
- **Tilt Out:** Sudden or sustained tilt change

The remaining options are configured in register **0x3B**, which contains the following interrupts:

- **Tap Out**: Single or double tap detected
- **I3C Out:** In-band interrupt protocol activity
- **Err Status Out:** Internal or communication error detected
- **Temp Drdy Out:** New temperature measurement available
- **Gyr Drdy Out:** New gyroscope measurement available
- **Acc Drdy Out:** New accelerometer measurement available
- **Fifo Watermark Out:** Configured FIFO memory threshold reached
- **Fifo Full Out:** FIFO memory completely filled

All interrupts can be mapped to either INT1 or INT2 by setting 0x1 for INT1 or 0x2 for INT2 in their respective register fields. Each physical interrupt can only be mapped to one source **at a time**. Attempting to map multiple interrupts **simultaneously** may cause measurement errors, physical pin output conflicts, or permanent damage to the sensor.

3.2 Board Topology



Top Side **Bottom Side**

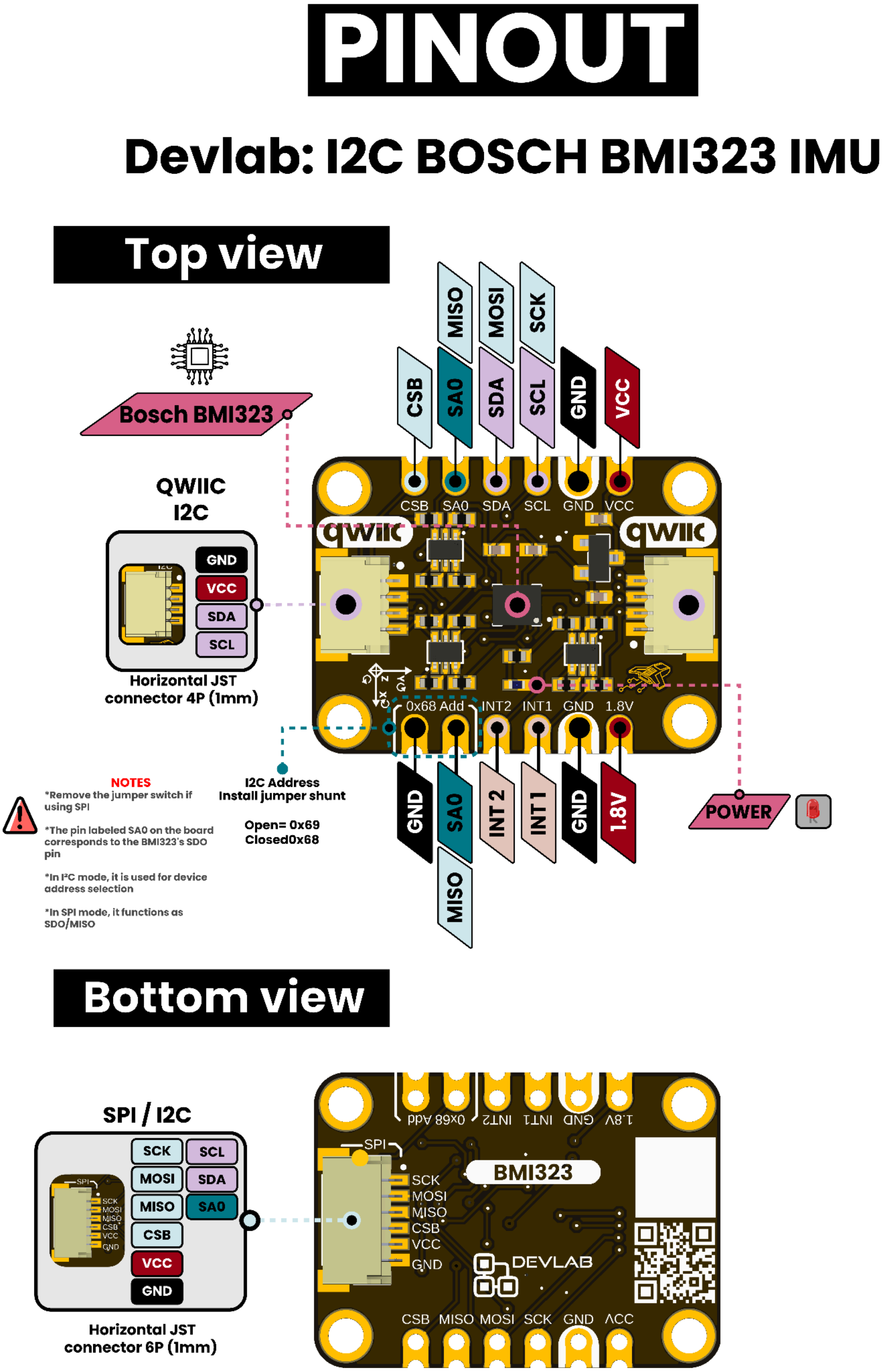
Views of Topology

Table 3.2.1 - Components Overview

Ref.	Description
IC1	Bosch BMI323 6-Axis Inertial Measurement Unit (IMU)
U1	ME6206A18XG 1.8 V Low Dropout (LDO) Voltage Regulator
Q1	BSS138AKDW Bidirectional Level Shifter for SCL/SCK and SDA/MOSI Signals
Q2	BSS138AKDW Bidirectional Level Shifter for CSB and SDO/SA0 (MISO) Signals
Q3	BSS138AKDW Bidirectional Level Shifter for INT1 and INT2 Signals
D1	Power On LED (Red)
J1	QWIIC Connector (JST SH 1 mm pitch) for I²C
J2	1×6 2.54 mm Castellated Header for Power and Communication Signals
J3	QWIIC Connector (JST SH 1 mm pitch) for I²C
J4	1×6 2.54 mm Castellated Header for Interrupt and Auxiliary Signals
J7	6-Pin JST SH 1 mm Connector for SPI Interface

4 Connectors & Pinouts

4.1 General Pinout



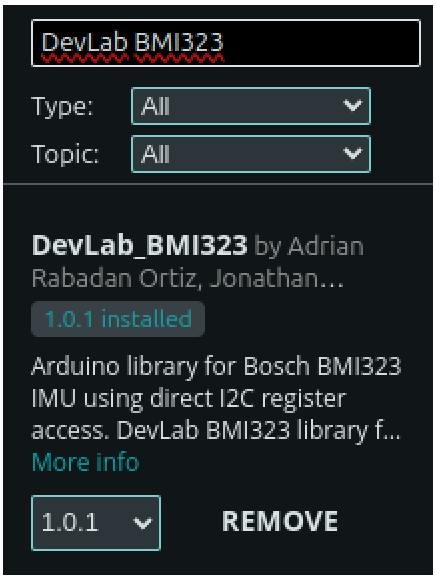
BMI323 IMU Sensor Module General Pinout

4.2 Pinout General Description

Pin	Label	Description	Typical Use
1	**VCC**	Main power supply input for the module. The onboard regulator generates the 1.8 V rail required by the BMI323.	Connect to the host power supply.
2	**GND**	Common ground reference for power and communication signals.	Connect to the host system ground.
3	**SCL / SCK**	Serial clock signal. Operates as **SCL** in I²C mode and **SCK** in SPI mode.	Connect to host I²C SCL or SPI SCK.
4	**SDA / MOSI**	Serial data signal. Operates as **SDA** for bidirectional I²C communication and **MOSI** in SPI mode.	Connect to host I²C SDA or SPI MOSI.
5	**SA0 / MISO**	Multifunction signal. In I²C mode, **SA0** selects the device address. In SPI mode, it operates as **SDO/MISO**.	I²C address selection or SPI MISO connection.
6	**CSB**	Chip-select signal used for SPI communication.	Connect to the host SPI chip-select GPIO when using SPI.
7	**INT1**	Programmable interrupt output from the BMI323.	Connect to an interrupt-capable host GPIO.
8	**INT2**	Second programmable interrupt output from the BMI323.	Connect to an interrupt-capable host GPIO when an additional interrupt output is required.
9	**1V8**	Regulated 1.8 V rail used internally by the BMI323.	Primarily for testing or voltage reference; avoid using it to power external loads.

5 Board Operation

The DevLab BMI323 module can be easily integrated into Arduino-compatible development boards through the I²C or SPI communication interface. Before running any application, install the DevLab BMI323 Arduino library using the Arduino Library Manager or download it from the official GitHub repository.



Hardware Interface

I2C Connection

For this example we use the UNIT Pulsar ESP32 C6, and this must be connect the BMI323 module to your board using I²C:

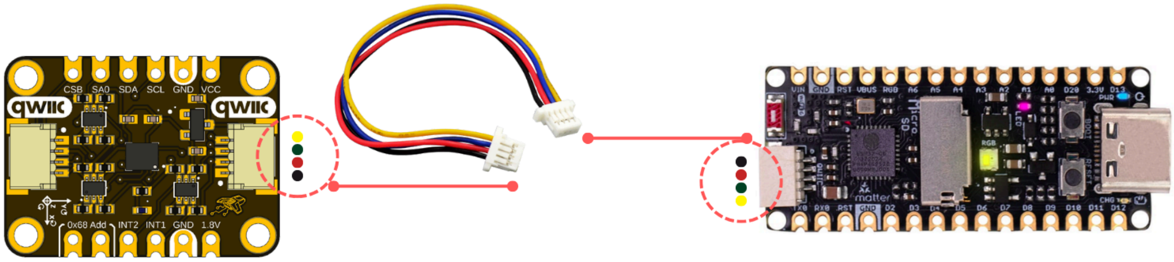
Signal	Description
VCC	Power supply
GND	Ground
SDA	6
SCL	7

In this example, the I²C pins are defined as:

```
#define SDA_PIN 6
#define SCL_PIN 7
```

Default I2C Address

Device	Address
BMI323	0x69



Example Operation

The example configures the sensor with:

Parameter	Value
Mode	High Performance
Average	No averaging
Filtering	ODR/4
Range	+/- Accelerometer 2G, +/-125 dps
ODR	800 Hz

The sensor reads three values:

Value	Description
Raw Data Accelerometer	Raw accelerometer data for each axis
Raw Data Gyroscope	Raw gyroscope data for each axis
Temperature	Internal temperature

After uploading the code, open the Serial Monitor at 115200 baud. If the module is connected correctly, the message `BMI323 initialized successfully` will appear, followed by continuous imu readings.

Basic Operation

1. Connect the BMI323 module to the microcontroller using the I²C or SPI interface.
2. Open the Arduino IDE.
3. Install the DevLab BMI323 library.
4. Open the Basic I2C example sketch provided with the library.
5. Select the correct board and serial port.
6. Compile and upload the example.
7. Open the Serial Monitor (115200 baud by default) to display the accelerometer, gyroscope, and temperature measurements.

After initialization, the library automatically verifies communication with the sensor by reading the device identification register. If communication is successful, the BMI323 is configured with the default settings defined by the library, and continuous sensor data becomes available.

Developers can modify the example code to configure:

- Accelerometer measurement range
- Gyroscope measurement range
- Output Data Rate (ODR)
- Power modes
- Digital filters
- FIFO buffer
- Interrupt generation
- Motion detection features

The DevLab BMI323 Arduino library provides a high-level API that simplifies sensor configuration while still allowing access to low-level register operations for advanced applications.

Example Code

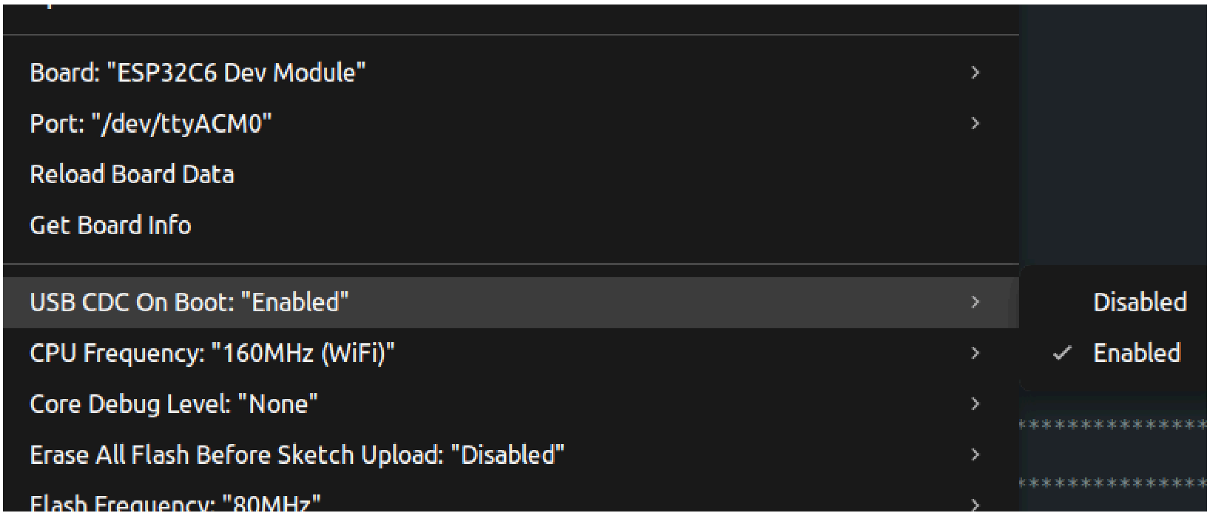
```
#include <DevLab_BMI323.h>
#define SDA_PIN 6
#define SCL_PIN 7
DevLab_BMI323 imu(Wire, 0x69);
BMI323_SensorData data;
void setup() {

 Serial.begin(115200);
 delay(1000);

 Serial.println();
 Serial.println("==================================================");
 Serial.println(" DevLab BMI323 - Basic Read Example");
 Serial.println("==================================================");
```

```
 Serial.println("Initializing BMI323...");

 // Initialize BMI323
 if (!imu.begin(SDA_PIN, SCL_PIN, 400000)) {

   Serial.println("ERROR: BMI323 initialization failed.");

   while (1) {
     delay(1000);
   }
 }

 Serial.println("BMI323 initialized successfully.");
 Serial.println();
}
void loop() {

 // Read sensor data
 if (imu.readData(data)) {

   Serial.println("------------------------------------------------");

   // Accelerometer data
   Serial.print("Accelerometer [raw]");
   Serial.print("  X: ");
   Serial.print(data.accX);

   Serial.print("   Y: ");
   Serial.print(data.accY);

   Serial.print("   Z: ");
   Serial.println(data.accZ);

   // Gyroscope data
   Serial.print("Gyroscope     [raw]");
   Serial.print("  X: ");
   Serial.print(data.gyrX);

   Serial.print("   Y: ");
   Serial.print(data.gyrY);

   Serial.print("   Z: ");
   Serial.println(data.gyrZ);

   // Temperature data
```

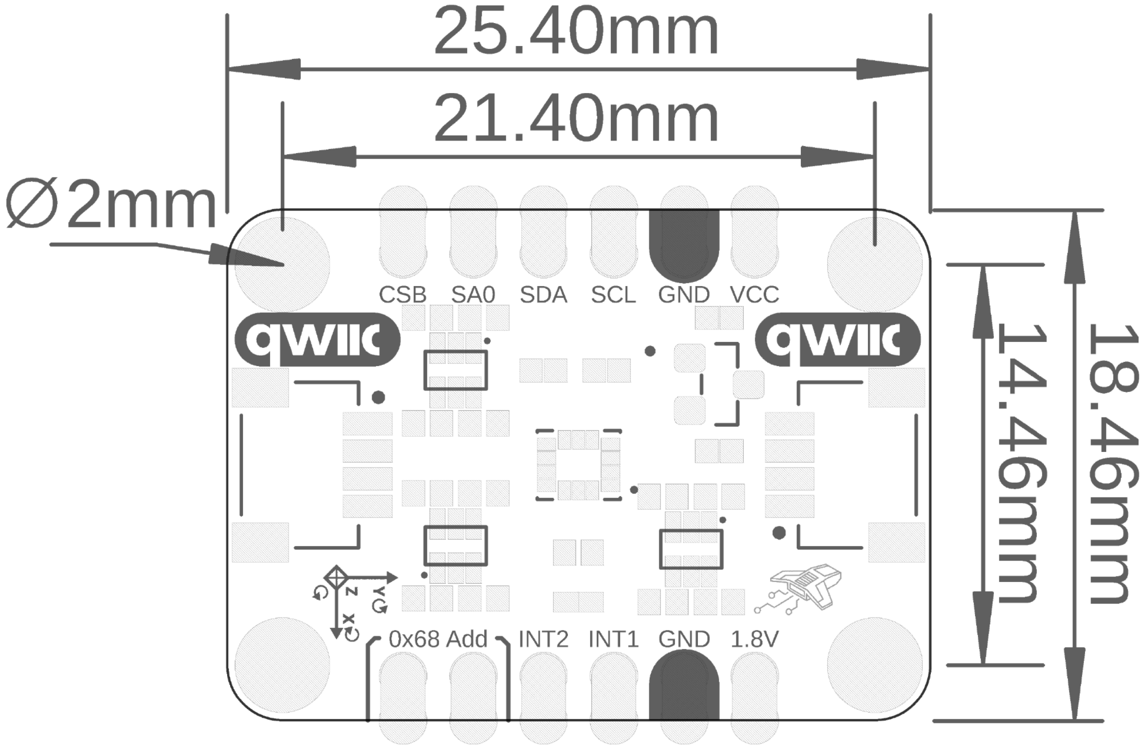
```
  Serial.print("Temperature   [C]");
  Serial.print("    ");
  Serial.println(data.temperatureC, 2);

} else {

  Serial.println("ERROR: Failed to read BMI323 data.");
}

delay(200);
}
```

Important

Make sure **USB CDC On Boot** is set to **Enabled**. This allows the ESP32-C6 to expose the USB Serial interface after boot, so the `Serial` output can be viewed in the Arduino Serial Monitor. If this option is disabled, the sketch will still run, but you may not see any serial messages.



6. Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

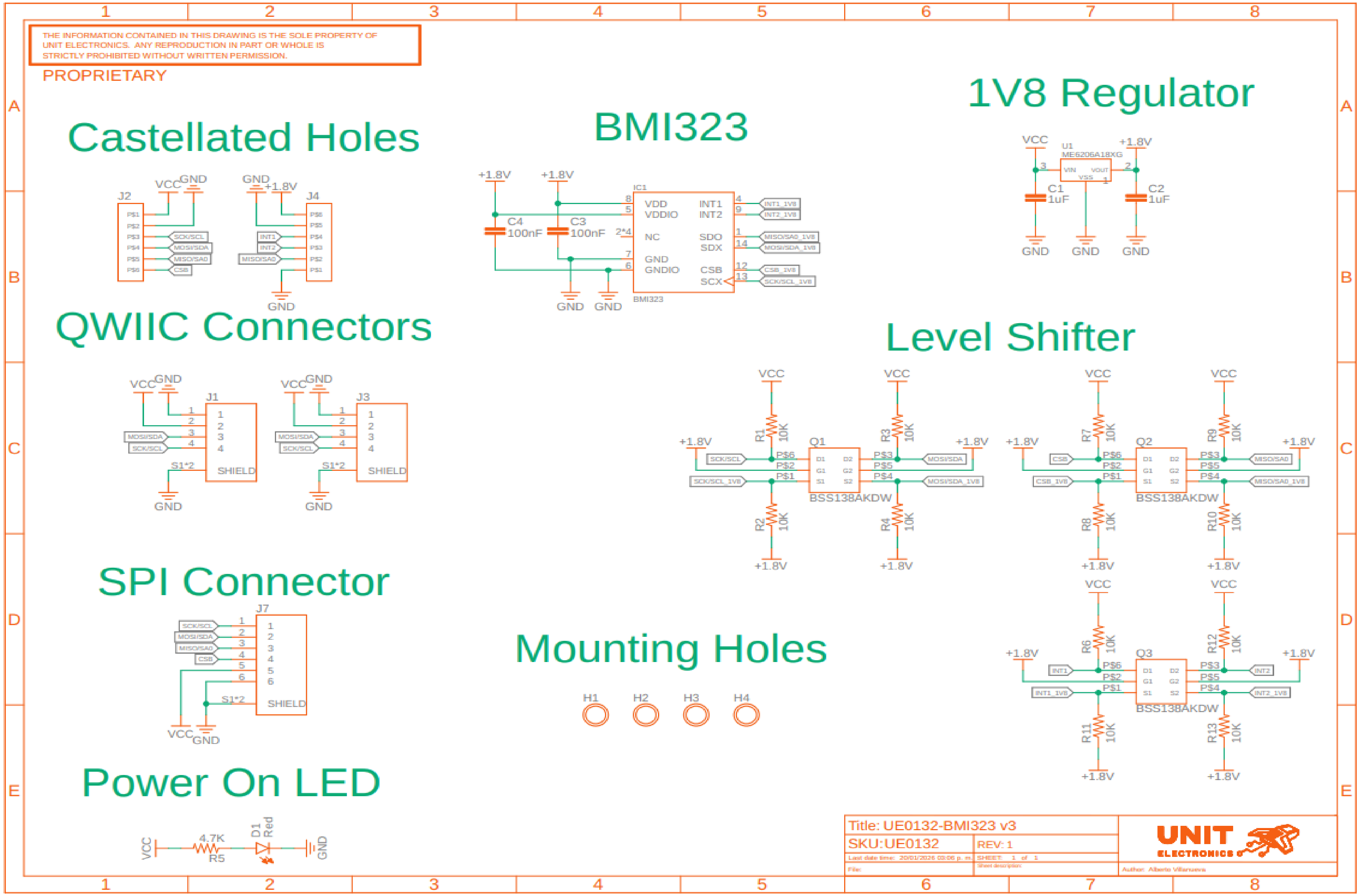
Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	**Link**
Documentation	https://wiki.uelectronics.com/wiki/devlab-i2c-bosch-bmi323-imu
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bosch_bmi323_imu
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download
DevLab BMI323 Arduino Library	https://github.com/UNIT-Electronics-MX/unit_library_devlab_bmi323

9 Appendix

9.1 Schematic
(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bosch_bmi323_imu/blob/main/hardware/unit_sch_v_1_0_0_0_ue0132_bmi323.pdf)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PROPRIETARY
Castellated Holes
BMI323
1V8 Regulator
QWIIC Connectors
Level Shifter
SPI Connector
Mounting Holes
Power On LED
U1
ME6206A18XG
C1
1uF
C2
1uF
C4
100nF
C3
100nF
IC1
VDD
VDDIO
NC
GND
GNDIO
INT1
INT2
SDO
SDX
CSB
SCX
BSS138AKDW
H1
H2
H3
H4
4.7K
R5
D1
Red
Title: UE0132-BMI323 v3
SKU:UE0132
REV: 1
UNIT ELECTRONICS